

TIDES Roadmap: 2026–2027+

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Note: [TIDES Program Management Gantt Chart](#) is a corresponding visualization of Roadmap's priorities.

About This Document

This is the TIDES Board's strategic roadmap. It describes the direction, priorities, and milestones for the Transit Integrated Data Exchange Specification (TIDES) through 2026, 2027 and beyond, spanning specification development, tooling, community growth, and adoption.

What this roadmap is. A strategic framework that the Board owns and uses to guide prioritization decisions over the next 18 months. It sets direction and identifies the major initiatives the stewardship team and community will pursue.

What this roadmap is not. A detailed execution plan. A companion implementation document held by the stewardship team provides the operational detail behind each section.

Funding posture. This document identifies which elements of the Roadmap are funded based on available resources at the time it was adopted. MobilityData Bridge funding sustains TIDES community management at a baseline level through July 2026, with capacity reduced to approximately two hours per week in July. All other workstreams in this roadmap, including v2.0 release work, the NTD reporting pilot framework, tooling, documentation expansion, and the agency survey, depend on additional funding that has not yet been secured. The stewardship team is actively exploring sources; beyond baseline community management, nothing in this roadmap should be read as a MobilityData commitment to deliver.

Strategic Principles

Every item on this roadmap should be evaluated against these principles. They operate alongside the principles already in the [TIDES Change Management Policy](#), which governs how specification changes are proposed, prioritized, and adopted.

1. **Outcomes first.** TIDES is driven by desired outcomes (clean, sustainable operational data; automated NTD reporting; new transit analytics capabilities), not by development of the specification as its own goal.
2. **Community-driven.** Priorities emerge from community input (structured interviews, the agency survey, working groups, GitHub issues), not top-down prescription.
3. **Vendor neutrality.** All engagement remains vendor-neutral. The goal is to bring vendors into the community, not position against them.
4. **Scope conservatism.** The roadmap reflects available resources and distinguishes between work that is committed and in development versus work that is aspirational and longer-term.

Near-Term Milestones (2026)

The following workstreams form the near-term backbone of TIDES delivery in 2026. Resourcing status for each item is tracked in the companion Gantt chart, published alongside this Roadmap and updated as resourcing evolves.

Workstream	Target
TIDES v2.0 release (breaking changes consolidated, PRs merged)	August 2026 (timed to APTAtech)
TIDES Briefing (introductory training session, v1.0 tables and v2.0 preview)	June 2026
GTFS-RT to TIDES converter documentation published	2026
TIDES-to-NTD derivation documentation	2026
Procurement support resources finalized for v2.0 publication	August 2026
Feature development governance: initial draft for Board comment	2026
Cross-standard coordination plan (TODS, GOFS, GTFS, OpRa)	2026
Sample data and sandbox tooling, initial release	2026
Code publishing in tides-implementations repository	Ongoing

Additional workstreams described throughout this document, including the agency survey, the NTD reporting pilot framework, and the open-source tools catalog, are in development. The stewardship team is actively exploring funding for these.

1. Core Specification Development

1.1 Version 2.0 Completion and Release

Target: August 2026, timed to APTAtech.

TIDES v2.0 bundles all breaking changes from the Fall 2025 Issues Working Group and incorporates lessons from the initial TIDES implementation at WMATA. This is the version agencies and vendors should target in their next ITS procurement. Once agencies begin requiring TIDES v2.0 compliance in RFPs, v2.0 becomes the market standard.

The remaining open issues targeted for this release (device_events rename, device_status table, snake_case standardization across 12 fields) are being resolved. Implementer interviews are running in parallel to verify that current implementers can support the v2.0 release target.

1.2 Release Strategy

The v2.0 launch will be a coordinated multi-channel announcement centered on APTAtech in August 2026, with groundwork laid through earlier 2026 community engagement. Procurement language will be published simultaneously, because agencies need the specification and the contract language at the same time to act on it. Direct outreach to implementing agencies and engaged vendors will precede the public announcement.

1.3 Future Releases (v2.1+)

Following v2.0, the community will scope v2.1 additions. Candidates include linked trips and passenger journeys (Issue #243, non-breaking) and aggregated ridership tables (Issue #244). A gap identification program of structured interviews with implementing agencies will surface additional needs.

Release cadence will follow the existing TIDES Release Management Policy: no more than one major release per year, with minor releases possible as the community converges on additional non-breaking content.

1.4 Candidate Features (Exploration Phase)

Before the community invests in any feature, two things must be in place: a formal feature development governance process (see Section 9.3) and clear prioritization criteria. All candidates below are at the **exploration phase**. They are not committed deliverables.

Tier 1: strongest community signal and clearest data gap.

- **Demand-Responsive Transit (DRT).** The widest data standardization gap in the community. Fixed-route service has GTFS and TIDES. DRT operational data remains largely unstandardized across vendor platforms.

Tier 2: strong interest, needs further scoping.

- Vehicle Monitoring System (VMS) data
- Equipment Monitoring data (elevators, escalators, fare gates; accessibility metrics)
- Real-time Prediction Accuracy and Availability (GTFS-RT prediction quality against TIDES actuals)
- Transit Signal Priority (TSP) operational data
- Crash and incident data
- Automated stop announcement data

Other candidates that may be core v2.x specification additions rather than separate features:

- Linked trips and passenger journeys
- Aggregated ridership tables
- Street segment and network matching, linear referencing
- Rail track circuit data
- Multi-fund fares and complex fare policies

2. NTD Reporting Initiative and Federal Engagement

NTD alignment has been identified as an important opportunity across every stakeholder conversation. It makes TIDES tangible to agencies, vendors, and FTA alike.

2.1 FTA Engagement

The stewardship team is engaged in ongoing dialogue with FTA on how TIDES can support NTD reporting modernization, family-friendly fare reporting, and broader transit data infrastructure goals. FTA has expressed interest in continued engagement around these topics, and follow-up discussions are underway. FTA input on the agency survey design is also being sought.

TIDES's momentum with FTA is real. The commitment at this stage is to explore and pilot, not yet a formal program.

2.2 NTD Reporting Pilot Framework

An NTD reporting pilot framework is in development to demonstrate TIDES-based automated NTD reporting. The pilot model would run TIDES-based NTD reporting in parallel with an agency's existing NTD approach, processing the same source data and producing monthly reporting output validated against the official submission.

The framework is being developed in coordination with several transit agencies, including WMATA (the first major TIDES implementation and the agency that shaped v2.0), Mountain Line in Missoula (a small-urban participant), and additional agencies are in discussion. Agency-level pilot work would produce documented methodology and reusable tools, which would in turn become the basis for a multi-agency pilot framework.

Target for initial pilot operations: mid-2026, contingent on participating agencies' fiscal year cycles and on funding arrangements.

2.3 Agency Survey

The agency survey generates intelligence for the TIDES community on adoption readiness and barriers, while providing data that helps FTA understand what NTD reporters are experiencing. The survey instrument is substantially complete, with methodology developed in consultation with UCLA ITS. FTA input on question design is being sought before distribution.

Survey outputs will feed specification development priorities, feature sequencing, adoption support targeting, and pilot cohort recruitment.

Survey distribution and analysis is a near-term priority pending funding.

2.4 Multi-Agency Pilot Framework

Building on lessons from initial agency pilots, a multi-agency pilot framework would extend TIDES-based NTD reporting to a diverse cohort of agencies. The framework's deliverables would include guidance documentation, a reference architecture, and sample/reference code to facilitate development of NTD data pipelines on TIDES at participating transit agencies, along with implementation patterns and lessons learned that the broader community can reference. The stewardship team is developing proposed selection criteria. Diversity goals include geographic, modal, size, and operator-type representation (including small and rural agencies). FTA engagement on the broader pilot is underway.

3. Documentation

Documentation is the bridge between specification and adoption.

3.1 TIDES Briefing

A structured training session covering v1.0 tables already in production and a v2.0 preview is planned for June 2026, funded by Cal-ITP. The session is designed for implementing agencies, vendor teams, and community members who need a grounded orientation to TIDES structure and use cases. Recordings and materials will be retained for ongoing community reference.

3.2 Foundation Documentation

The foundation documentation suite, developed through the WMATA TIDES implementation, is being published to the TIDES site:

Implementation Guide (starting sections): Foundational guidance for transforming data into the tables that agencies implement first; building production TIDES pipelines at any scale.

Architecture Framework: How TIDES fits into an agency's data infrastructure; event versus summary data patterns; GTFS and GTFS-RT integration; the maturity ladder from laptop to enterprise.

Open-source dbt Transformation Packages: Open-source reusable transformation patterns from the initial TIDES implementation at WMATA. The packages are published in the tides-implementations repository on the TIDES GitHub organization, under Apache 2.0. The model is "build once, use everywhere."

3.3 Advanced Documentation

A second wave of documentation is planned to follow the release of v2.0 of the TIDES specification:

Implementation Guide for advanced topics, including narrowly focused issues and new tables in v2.0.

v1.0 to v2.0 migration guide (critical given breaking changes).

TIDES-to-NTD derivation documentation, connecting specification to reporting requirements.

4. Procurement Support

Procurement resources may be the single most impactful adoption mechanism. If TIDES is not in the RFP, it does not happen. The GTFS parallel is instructive: GTFS became universal because agencies required it in contracts and vendors adapted.

4.1 Procurement Resources

The resources serve three distinct procurement scenarios:

1. Source-system procurement (CAD/AVL, APC, AFC, fare systems, and similar) — TIDES requirements specify the format and field-level expectations for the data the system must produce.
2. System implementation, integration, and data analysis/reporting services procurement — TIDES requirements specify the formats vendors must accept and produce as part of broader delivery scope.
3. TIDES implementation services specifically — procurement of help getting an agency's data into TIDES.

The full resource suite is planned for publication simultaneously with v2.0 at APTAtech in August 2026:

- Procurement Language and Requirements Template
- Definitions Appendix
- Translation Guide
- Due Diligence Checklist
- Compliance Verification Procedures
- Sample RFP Language inserts
- One-page Factsheet for agency procurement offices

Both the Procurement Language and Requirements Template, v1.0 documents (January 2026), have been reviewed by an initial pilot agency and the TIDES Board. Feedback from agency, vendor, and Cal-ITP procurement reviewers is being incorporated into an updated version targeted for publication alongside v2.0 in August 2026.

4.2 Vendor Engagement Through Procurement

When agencies include TIDES requirements in RFPs, vendors respond. Key vendor-facing concepts:

“TIDES Ready” tier. Vendor data that maps 1:1 to TIDES fields but is not yet in TIDES format. This is a pragmatic starting point that gives agencies leverage.

Vendor field mapping as a deliverable. Vendors document how their system maps to TIDES fields, giving agencies the transparency to own conversion.

Outcome: a growing share of agency RFPs requiring TIDES, an expanding set of vendors offering TIDES output as a standard product capability, and over time, transit agencies that can procure TIDES-capable systems off the shelf rather than commission custom builds.

Just as importantly, the vendor pillar matures into active collaboration on the standard's evolution: vendors implementing TIDES against real production data surface edge cases that sharpen the spec, publish open-source reference connectors that compress adoption cost for the next implementer, and bring forward feature proposals for underserved areas through Working Group participation.

5. Tools Development

GTFS did not succeed because the specification was well-designed. It succeeded because of the ecosystem of consumers, validators, converters, and tools built around it. TIDES needs the same.

5.1 Short-Term Priorities

Priority 1: TIDES-to-NTD Reporting Pipeline. Within the operations-data subset of NTD reporting (not a complete NTD solution), every U.S. transit agency must submit NTD reports. Producing the ridership (UPT, PMT) and service (VRM, VRH, VOMS) metrics today is a time-consuming reconciliation across CAD/AVL exports, APC summaries, and fare system records. A TIDES-based pipeline automates that derivation and lays the foundation for emerging FTA priority metrics around on-time performance and service reliability. The NTD reporting pilot framework (see Section 2.2) will produce the first reference implementation; what pilot agencies build will become replicable for every agency that follows.

Priority 2: Sandbox and Development Environment. Someone wanting to build a TIDES tool today faces a cold-start problem. Almost no sample data exists, and real agency data, particularly fare transaction data, contains PII that makes open sharing essentially impossible regardless of sanitization. A synthetic data generator seeded from GTFS, GTFS-RT (or CAD/AVL), APC, and AFC data produces plausible test data without those constraints. The second component is a CodeSpaces-style starter environment (forkable, pre-configured with DuckDB, dbt, and synthetic data) that eliminates the setup barrier entirely.

Priority 3: Shared library of open-source transformation packages. The hardest part of a TIDES implementation is the first time: figuring out how to map vendor source-system data outputs to TIDES fields. WMATA's Smart Data Hub dbt packages are the founding contribution to this library, now published in the [tides-implementations](https://github.com/mobilitydata/tides-implementations) repository under Apache 2.0. The model going forward is “build once, use everywhere”: each new agency implementation contributes packages back, expanding coverage of vendor source systems over time.

Priority 4: GTFS-RT to TIDES Converters. Two GTFS-RT to TIDES converters exist today, both community-developed. For agencies already archiving GTFS-RT vehicle position data, which is most agencies with modern CAD/AVL, these converters provide a built-in path to TIDES vehicle_locations with

minimal additional work. Refining, documenting, and publishing them as ready-to-use tools is a 2026 deliverable, and demonstrates that TIDES adoption can start from infrastructure agencies already have.

Priority 5: Operational Data Validation. Data quality tooling is what makes TIDES trustworthy for NTD reporting and other high-stakes use cases. Agencies need a clear signal that their data is valid before it enters any reporting pipeline. The Frictionless Data framework provides basic schema validation, and the near-term work adds transit operations-specific checks (negative passenger loads, timing inconsistencies, speed anomalies). A “GTFS Validator equivalent for operational data” is a concept the community has consistently rallied around. It gives procuring agencies a concrete acceptance criterion for vendor-produced TIDES data and allows agencies to more confidently integrate data from multiple operational sources.

5.2 Medium-Term Tools

Targeted for late 2026 through mid-2027:

- AVL-to-OTP / Speed / Reliability Pipeline
- Semantic data quality tools (e.g., missing data vs. no service, device health monitoring)
- Data Sharing Repository (public TIDES dataset registry, modeled on the GBFS systems catalog)
- Open data portal tooling
- Aggregation tools (an essential component of NTD reporting pipeline work)
- Conformance testing for vendor implementations

5.3 Long-Term Tools

Aspirational, contingent on community co-investment and funding:

- Demand-response transit (DRT) feature tooling (dependent on Board-sanctioned feature development; see Section 9.3)
- Passenger journey reconstruction (linked trips)
- Performance Dashboard in a Box
- Common API interface standard
- Research data access tools and APIs

6. Community Development

6.1 Community of Practice

The Contributors Group (specification-focused, monthly) and the Implementers Group (implementation-focused, monthly) are the backbone of the TIDES community. Ongoing facilitation, agenda-setting, and follow-up are stewardship team responsibilities.

A structured interview program is planned to supplement community meeting discussions. It uses a lightweight protocol with implementing agencies and community members to surface needs that do not emerge through GitHub issues or group meetings. Dual purpose: specification gap identification and adoption intelligence. Implementer interviews are the near-term focus, verifying readiness of current implementers ahead of v2.0.

6.2 Vendor Engagement

Vendor engagement is an elevated priority as v2.0 approaches and procurement requirements appear in active RFPs. Vendors caught off-guard by TIDES requirements in bids are a risk to adoption. Vendors who are prepared become advocates.

The strategy is to involve vendors in procurement resource review, invite them to Implementers Group meetings, and offer early access to feature development working groups. All engagement remains vendor-neutral. The goal is community inclusion, not positioning.

6.3 Conference Strategy

The flagship industry event for TIDES in 2026 is **APTatech 2026 (August)**, where the v2.0 release and the procurement resources will be formally launched. Additional presentations and workshops are planned at other industry events throughout the year, including data practitioner and rail audiences, with the strategic direction shifting from concept introduction to tool demonstrations.

TIDES maintains a presence at relevant industry conferences and events to introduce the specification and surface lessons from implementing agencies. Specific event participation is planned on a rolling basis with the community.

6.4 Academic and Research Partnerships

Active collaborations with UCLA ITS (survey methodology, linked trips) anchor the academic side, with potential connections to MIT Transit Lab (ROVE tool, TIDES-formatted research data), Imperial College London (benchmarking), and Oregon State (historical TCRP program involvement).

6.5 International Coordination

OpRa (CEN Standard, EU). OpRa and TIDES address the same fundamental need, standardizing data management for historical operational transit data, but for different regulatory contexts. The two efforts are in active coordination through their respective technical committees. OpRa's Technical Specification is still in development, creating a window for proactive interoperability design.

7. Implementation Support and Agency Adoption

Implementation support bridges the gap between “TIDES exists as a specification” and “TIDES is used in the real world.”

Scalable implementation models:

Entry level (analyst on a laptop). Sandbox environment, synthetic data, getting-started documentation.

Mid-tier (shared server, modest IT support). Implementation guide, reference dbt packages, community working group participation.

Enterprise. Reference code from the [tides-implementations repository](#), direct consultation, co-development of community tools.

Potential support models

- Open-source reference implementations in the [tides-implementations repository](#).
- Implementation documentation on [tides-transit.org](#), including the Implementation Guide, Architecture Framework, and migration materials.
- Procurement support resources, including RFP language, requirements templates, and a Procurement Language Factsheet.
- Community of practice through the monthly Contributors and Implementers Group meetings, the TIDES Slack workspace, and GitHub Discussions.
- Community-contributed tooling, including open-source validators, GTFS-RT to TIDES converters, and sandbox environments.
- Public briefings and conference presence that introduce the spec and surface lessons from agencies actively implementing.

Key maturity milestone: code from one agency deployed at another agency. This is the signal that TIDES has transitioned from a specification exercise to a shared operational resource.

The agency survey will generate targeting intelligence, identifying where adoption barriers are highest and where support resources can have the most impact.

8. Cross-Standard Coordination

TIDES is designed to complement, not compete with, the broader transit data standards ecosystem.

GTFS ecosystem. TIDES integrates through common identifiers (trip_id, stop_id, route_id). Maintaining this compatibility as both standards evolve is essential.

TODS (Transit Operational Data Standard). Complementary. TODS extends GTFS for scheduled operations; TIDES covers observed and historical data. Joint procurement resource development is underway.

GOFS (General On-Demand Feed Specification). Rider-facing on-demand information. A TIDES DRT feature would cover the operational layer. Coordination ensures compatibility.

OpRa, Transmodel, NeTEx. See Section 6.5.

A formal cross-standard coordination plan, mapping TIDES's relationship to each standard and identifying translation and governance coordination opportunities, is a 2026 deliverable.

9. Governance

The TIDES governance model has functioned well through the v1.0 release and v2.0 development. The roadmap identifies the following items for formalization and Board guidance.

9.1 Core Governance Updates

Board role in releases. Two or more Board members are required for merges to main. This should be formally documented in the governance docs.

Community meeting cadence. The Contributors Group and the Implementers Group meet monthly. Facilitation and agenda-setting by the stewardship team is an ongoing maintenance function.

Change management. The change proposal and review process needs to be more formally documented and published, for new contributors and for vendors who need to understand how the specification evolves. The existing TIDES Change Management Policy is the foundation.

Scaling governance. As the community grows (more agencies, more vendors, more use cases), current governance structures may need to evolve. The stewardship team will monitor adequacy.

9.2 Standing Release Cadence

TIDES needs a sustainable, predictable release cadence that the community can build against. The GTFS ecosystem provides precedent: GTFS uses consensus-based voting with implementation requirements before voting, and GBFS uses a Release Candidate model. The TIDES Release Management Policy already provides the framework (no more than one major release per year, with minor releases as content

converges); the open question is how to operationalize it for a community of TIDES's current size and Board governance model.

9.3 Feature Development Governance

TIDES is establishing a clear process for how new features are proposed, evaluated, prioritized, and adopted into the specification. This formalizes what has been an informal practice and is a prerequisite for transparent feature development going forward. All feature development happens through Board-sanctioned working groups; TIDES does not run a parallel third-party process.

Proposed six-phase process:

1. **Exploration.** Community identifies a need through structured interviews, GitHub issues, or conference discussions.
2. **Initial assessment.** Stewardship team assesses technical feasibility, overlap with existing tables, alignment with TIDES principles, and relationship to other standards. The assessment is documented and shared publicly.
3. **Community input and Board decision.** The Board makes a formal decision on whether to prioritize the feature. This is the key governance moment. It prevents the stewardship team from pursuing features without Board mandate.
4. **Working group formation.** Relevant agencies, vendors, and subject matter experts convene. A draft schema is developed through an iterative community process.
5. **Prototype implementation and validation.** At least one agency or vendor implements the draft feature and validates it against real data before any vote.
6. **Formal adoption.** Community review period, Board vote, merge into the specification.

This process will be formalized in TIDES governance documentation and operationalized through the quarterly prioritization cycle defined in the Change Management Policy.

10. Ongoing Maintenance

The following are ongoing operational responsibilities of the stewardship team. They are not discrete deliverables but essential functions that require sustained effort.

Maintenance Item	Cadence
Contributors Group facilitation (agenda, hosting, follow-up)	Monthly
Implementers Group facilitation (agenda, hosting, follow-up)	Monthly

Maintenance Item	Cadence
GitHub repository management (issue triage, PR review and merges, governance admin)	Continuous
Communications and stakeholder management (mailing lists, Slack, agency and vendor inquiries)	Continuous
Website and online presence (tides-transit.org updates, resource inventory)	Continuous
Progress reporting	Monthly
Annual roadmap updates	Annual
FTA relationship management and coordination	Ongoing
Documentation maintenance (keeping published docs current as specification evolves)	Ongoing